



CS 418: Introduction to Data Science

Syllabus

Spring 2026

Logistics, Staff, and Contacts

Lectures

Days: Mondays and Wednesdays
Time: 12:30PM - 01:45PM
Location: LC C1
CRN: 43592 (Undergraduate), 48632 (Graduate)

Instructor

Gonzalo Bello

Email: gabello@uic.edu
Office: CDRLC 4417
Office Hours: Mondays and Wednesdays, 02:00PM - 02:50PM

Meetings outside office hours are available by appointment.

Teaching Assistant

Yuqing Liu

Email: yliu363@uic.edu
Office: CDRLC 2404
Office Hours: Tuesdays and Thursdays, 01:00PM - 02:00PM

Course Website

Blackboard: uic.blackboard.com/ultra/courses/_303030_1/cl/outline

Piazza: piazza.com/uic/spring2026/cs418mw1230pm

Gradescope: gradescope.com/courses/1221866 (you will be automatically added to *Gradescope* before the first assignment)

All grades and course materials (lecture slides, lecture videos, quizzes, homework assignments, lab assignments, projects, etc.) will be posted on *Blackboard*.

All course communication will occur on Piazza (see *Piazza* section for details).

Course Description

Provides an in-depth overview of data science in engineering. Topics include modeling, storage, manipulation, integration, classification, analysis, visualization, information extraction, and big data in the engineering domain. *Course Information:* 3 undergraduate hours. 4 graduate hours. Extensive computer use required.

The topics covered in each lecture are shown in the course schedule.

Note: CDRLC is the new CS building (Computer Design Research and Learning Center)



*"The ability to manipulate and understand data is increasingly critical to discovery and innovation. As a result, we see the emergence of a new field—**data science**—that focuses on the processes and systems that enable us to extract knowledge or insight from data in various forms and translate it into action.... [D]ata science has evolved as an interdisciplinary field that integrates approaches from such data-analysis fields as statistics, data mining, and predictive analytics and incorporates advances in scalable computing and data management."*

Berman, F., Rutenbar, R., Hailpern, B., et al., [Realizing the Potential of Data Science](#)

Course Objectives

Upon successful completion of this course, you should be able to:

- **Collect** and **integrate** data from multiple sources.
- **Prepare** data for analysis by detecting and correcting data quality problems and transforming data into an appropriate format.
- **Visualize** data to convey information to the viewer, such as patterns or trends.
- **Explore** data through summary statistics and visualization to understand and summarize its underlying structure.
- **Design** and **build** data models for analysis tasks, such as regression, classification, and clustering.
- **Evaluate** the results of data models, **analyze** their statistical significance, and **diagnose** their limitations, such as underfitting or overfitting.
- **Communicate** data analysis results through written reports and oral presentations.
- **Recognize** the advantages and disadvantages of existing frameworks for data storage and processing.
- **Understand** the ethical guidelines for the use of data and statistics.

Prerequisites

- CS 251: Data Structures.
- STAT 381: Applied Statistical Methods I, IE 342: Probability and Statistics for Engineers, or ECE 341: Probability and Random Processes for Engineers.

Textbook

No textbook is required.

Piazza

All course communication will occur on Piazza. This includes announcements from the instructor and TA and questions from students. **Please don't email the teaching staff directly.**

You are strongly encouraged to participate actively on *Piazza*. The teaching staff will monitor the message boards and endorse correct answers from students. **Posts unrelated to the course are not allowed and will be deleted.**

When posting, you should follow these guidelines:

- *Search before posting*; your question may have already been answered.



- Use an appropriate tag (e.g., hw1, project1, exam, lecture).
- Post publicly when possible so other students can benefit from your question. Use private messages only for grading issues or posts of a personal nature. You may post anonymously to other students (but the teaching staff will still see your name).
- **Don't post answers to assignment questions**; instead, offer hints or clarifications of the material.

iClicker

We will use [iClicker](#) for class participation. To join the course in *iClicker*, please follow the instructions below.

- Go to join.iclicker.com/VLJV.
- Create an *iClicker* account by entering your first name, last name, email, and NetID. **Make sure that you use your UIC email**. If you already have an *iClicker* account, you should just sign in.

To participate in the lectures, you must sign into your *iClicker* account from your computer or use the *iClicker Student* app from your phone. **You don't have to pay anything for this**.

We will start using *iClicker* in the second lecture (**Wednesday, January 14**).

Office Hours

We strongly encourage you to attend office hours throughout the semester to ask questions about the material or the assignments. Here is some advice to make the most out of office hours:

- *Go early and often!* Each topic relies on a strong understanding of what came before so you should make sure you understand all the material as it comes up.
- Even if you are not sure what questions to ask, *you can still go to office hours!* You may benefit from questions from other students and you can work through problems with the instructor or TA.
- *You don't have to arrive exactly at the start of the office hours*; you can arrive (and leave) at any time before it ends.

TA office hours will start in the second week of classes (from **Tuesday, January 20**).

Grading

The final grade will be determined by the following components:

Component	Quantity	Weight	Details
Class Participation	Every lecture	5%	Individual, in person
Online Quizzes	10*	5%	Individual, online
In-Class Quizzes	8*	15%	Individual, see dates in course schedule
Exam	1	20%	Individual, see dates in course schedule
Lab Assignments	6**/7***	15%	Individual
Homework Assignments	4**/5***	15%	Individual
Projects	2	15%	In groups
Final Project	1	10%	In groups



* The lowest quiz grade will be dropped at the end of the semester.

** For undergraduate students.

*** For graduate students.

The final letter grade will be determined using the following scale:

$90 \leq \mathbf{A} \leq 100$; $80 \leq \mathbf{B} < 90$; $70 \leq \mathbf{C} < 80$; $60 \leq \mathbf{D} < 70$; $0 \leq \mathbf{F} < 60$

To pass this course, you must achieve a grade of at least 40% on the exam. Otherwise, you will receive an **F** in the course regardless of your overall grade.

Course Policies

Attendance

Attendance to lectures is required. All lectures will be recorded and posted on *Blackboard*. You are responsible for making up any course material covered during missed classes.

Class Participation

Class participation will be recorded at every lecture. You must attend the lecture and answer the *iClicker* questions and lecture assignments to receive credit for that lecture. Answers for *iClicker* questions and lecture assignments will be graded on **completion**, not correctness.

Additionally, you may miss **up to 5 lectures** without any penalty. This drop policy is intended to cover absences from illness, sports, personal emergencies, etc. and will be applied automatically at the end of the semester.

Exam and In-Class Quizzes

The exam and in-class quizzes will be taken in person during class times. Dates for the exam and in-class quizzes are shown in the course schedule. There will be no make-up if you miss the exam or an in-class quiz. **The lowest in-class quiz grade will be dropped at the end of the semester.**

Online Quizzes

Online quizzes will be taken on *Blackboard*. Due dates for online quizzes are shown in the course schedule. **The deadline for the submission is Friday at 11:59PM (Central Time) on the week due. The lowest online quiz grade will be dropped at the end of the semester.**

Lab Assignments and Homework Assignments

Lab assignments must be submitted on *Gradescope*. Homework assignments must be submitted on *Blackboard*. Further instructions for the submission will be given in each assignment. Lab assignments and homework assignments are **individual**.

Due dates for lab assignments and homework assignments are shown in the course schedule. **The deadline for submission is Sunday at 11:59PM (Central Time) on the week due.** Any deadline extensions are up to the discretion of the instructor and will be announced to the class.

Graduate students are required to submit all **5 homework assignments** and **7 lab assignments**. Undergraduate students are required to submit only **4 homework assignments** and **6 lab assignments**.



If an undergraduate student submits all homework assignments or lab assignments, the **lowest one will be dropped at the end of the semester.**

Projects

Projects must be submitted on *Gradescope*. Further instructions for the submission will be given in each project. Projects must be completed in **groups of 3 undergraduate students** or **groups of 2 graduate students**.

Due dates for projects are shown in the course schedule. **The deadline for submission is Sunday at 11:59PM (Central Time) on the week due.** Any deadline extensions are up to the discretion of the instructor and will be announced to the class.

Late Submissions

Late submissions of assignments and projects will be accepted **within 0-12 hours after the deadline with a 5-point penalty** and **within 12-24 hours after the deadline with a 10-point penalty**. No late submissions (penalty or not) will be accepted more than 24 hours after the deadline.

No late submissions will be accepted for quizzes or the final project.

Regrade Requests

Regrade requests for lab assignments, projects, in-class quizzes, and the exam may be submitted on *Gradescope*. Regrade requests will **not** be accepted **more than one week** after grades for that assignment, project, quiz, or exam have been posted. To request a regrade, you must follow these guidelines:

- *Review the sample solution (on Blackboard) to understand how to solve the problem.*
- *Review the grading rubric (on Gradescope) to understand how points were assigned.*
- *Based on the sample solution and the grading rubric, write a clear and complete description explaining **why** you should receive more points in the problem.*

If you don't follow these guidelines, the teaching staff will dismiss your regrade request without further review.

Academic Honesty

All assignments, quizzes, and exams are individual. You are required to submit your own solutions and acknowledge any sources used.

Offering or receiving any kind of unauthorized or unacknowledged assistance (from students, friends, family, tutors, textbooks, generative AI tools, or the Internet) **is a violation of the University's academic integrity policies**, will result in a grade of zero for the assignment and will be subject to disciplinary action (see **Academic Integrity** section below).

Academic Integrity

As an academic community, UIC is committed to providing an environment in which research, learning, and scholarship can flourish and in which all endeavors are guided by academic and professional integrity. All members of the campus community—students, staff, faculty, and administrators—share the responsibility of



insuring that these standards are upheld so that such an environment exists. Instances of academic misconduct by students will be handled pursuant to the [Student Disciplinary Policy](#).

Disability Accommodations

UIC is committed to maintaining a barrier-free environment so that students with disabilities can fully access programs, courses, services, and activities. Students with disabilities who require accommodations for access to and/or participation in this course are welcome but must be registered with the Disability Resource Center (DRC). You may contact the DRC at 312-413-2183 (v) or 773-649-4535 (VP/Relay). For more information on UIC's policies on working with students with disabilities, please see the University's [Guide for Accommodating Students](#).

Religious Holidays

Students who wish to observe their religious holidays shall notify the faculty member by the tenth day of the semester of the date when they will be absent unless the religious holiday is observed on or before the tenth day of the semester. In such cases, the student shall notify the faculty member at least five days in advance of the date when they will be absent. The faculty member shall make every reasonable effort to honor the request, not penalize the student for missing the class, and if an examination or project is due during the absence, give the student an exam or assignment equivalent to the one completed by those students in attendance. If the student feels aggrieved, they may request remedy through the campus grievance procedure.

Mental Health

We value your mental health and emotional wellness as part of the UIC student experience. The UIC Counseling Center offers an array of services to provide additional support throughout your time at UIC, including workshops, peer support groups, counseling, self-help tools, and initial consultations to speak to a mental health counselor about your concerns. Please visit the [Counseling Center](#) website for more information. Further, if you think emotional concerns may be impacting your academic success, please contact your faculty and academic advisers to create a plan to stay on track.

Classroom Environment

This course will follow the [UIC Department of Computer Science Code of Conduct](#). Non-compliance will be reported to the Director of Undergraduate Studies, and possibly the Dean of Students and the Office for Access and Equity.

UIC is committed to the most fundamental principles of academic freedom, equality of opportunity, and human dignity involving students and employees. Freedom from discrimination is a foundation for all decision making at UIC. Students are encouraged to study the University's [Nondiscrimination Statement](#). Students are also urged to read the University's [Public Formal Grievance Procedures](#). Information on these policies and procedures is available on the website of the University's [Office of Access and Equity](#).

NOTE: The syllabus is subject to change.

Last Modified: January 16, 2026